

Muhammad Rizwan Ali

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SUMMARY

Full-stack software engineer (MERN) building AI-integrated web apps. Recently shipped a learning platform end to end - a RAG pipeline that grounds AI answers in students' own PDFs, real-time multiplayer on Socket.io, and a self-hosted engine running 15+ languages and carried it all the way to production.

EDUCATION

University of Sargodha

Bachelor of Science in Software Engineering - CGPA: 3.48 / 4.0

Sargodha, Pakistan

2022 - 2026

PROJECTS

JobPortal - Multi-Tenant Recruitment Platform

2026

Next.js, TypeScript, Node.js, Express, MongoDB, Redis, BullMQ

[Live Demo](#) | [GitHub](#)

- Built a recruitment platform supporting 3 platform roles, 4 company roles, and a 7-stage hiring pipeline covering job publishing, applications, interviews, hiring decisions, and moderation.
- Implemented verified-email authentication with 15-minute JWT access tokens, rotated and hashed refresh tokens in HttpOnly cookies, and company-scoped authorization.
- Protected application integrity through applicant snapshots, screening responses, MongoDB transactions for 1–10 job submissions, and a compound unique index preventing duplicate applications.
- Processed 4 asynchronous workflows using BullMQ workers with 5-job concurrency and 3-attempt exponential backoff, alongside Redis caching, distributed rate limiting, text search, and cursor pagination.

EduQuest - AI-Powered Gamified Learning Platform

2025 - 2026

React, Node.js, MongoDB, LangChain, Gemini, Socket.io

[Live Demo](#) | [GitHub](#)

- Built a Retrieval-Augmented Generation (RAG) pipeline with LangChain, Gemini embeddings, and MongoDB Vector Search, letting students get answers grounded in their own uploaded PDFs, with auto-generated quizzes and context-aware explanations.
- Engineered real-time multiplayer coding competitions with Socket.io, featuring live leaderboards and synchronized game state across players.
- Wired up Redux and TanStack Query for frontend state and data fetching, and built the Multer + Cloudinary upload flow that stores avatars and the student PDFs powering the RAG system.
- Implemented Google OAuth 2.0 and JWT authentication plus a self-hosted Piston code-execution service on Docker/AWS supporting 15+ programming languages.
- Shipped the full system end-to-end - schema, REST API, React frontend, and cloud deployment on Vercel and Render.

VideoTube - Backend API

2024

Node.js, Express.js, MongoDB, JWT, Cloudinary, Multer

[GitHub](#)

- Built a YouTube-style REST API with Node.js, Express.js, and MongoDB following MVC architecture.
- Implemented dual-token JWT authentication (access + refresh) with bcrypt hashing and httpOnly cookies for secure session management.
- Integrated Cloudinary and Multer for avatar and cover-image uploads via multipart/form-data.
- Wrote MongoDB aggregation pipelines for live subscriber counts, channel profiles and nested watch history.
- Created reusable asyncHandler, ApiError, and ApiResponse utilities for consistent error handling across all routes.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, HTML5, CSS3

Frontend: React.js, Next.js, Redux, TanStack Query, Tailwind CSS

Backend: Node.js, Express.js, REST API Design, Socket.io

Databases: MongoDB, MongoDB Vector Search, PostgreSQL

AI / LLM: LangChain, RAG Pipelines, LLM Integration

Auth & Cloud: JWT, Google OAuth 2.0, bcrypt, Cloudinary, Multer

Developer Tools: Git, GitHub, Docker, Postman

